

# Trading Strategies — Living Document

A running catalogue of every signal/strategy we've considered, built, tested, or shelved — with **status and evidence**, so we refine over time instead of re-litigating. Update the changelog at the bottom on every material change.

**Golden rule:** nothing earns risk until it survives **point-in-time features + friction-aware, regime-split backtesting**. The broad-sentiment edge looked great and died exactly this way — see `backtest/engine.py`.

## Status legend

|               | Meaning                                                                          |
|---------------|----------------------------------------------------------------------------------|
| ✔ Validated   | Survived friction-aware, regime-split backtest (still preliminary — see caveats) |
| ● Marginal    | Real effect, but fragile net of costs / not sign-stable                          |
| ✖ Failed      | Tested and does not work (kept here so we don't retry it)                        |
| 🧪 Untested    | Built and wired, awaiting data to evaluate                                       |
| 💡 Idea        | Backlog — not yet built                                                          |
| 🔄 In progress | Data/build task underway                                                         |

## Progress at a glance

Last updated: 2026-06-03. Update this section whenever a status, metric, or next-step changes.

## Strategy scoreboard

| Strategy                                                                            | Class         | Status                                                             | Headline result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Next action                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>retail_fear</code> (capitulation)                                             | Macro overlay | ✗ <b>FAILS the random-entry null</b>                               | Looked validated through excess/MT5/OOS/optimization — but vs <b>random entry</b> (same mechanics, random dates): <b>GOLD is 4th-percentile (WORSE than random; B&amp;H +240% crushes it)</b> , AUDJPY only 76th (weak, n.s.). The apparent edge was <b>gold's drift</b> , not timing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>do not deploy.</b> Re-examine only if a turn-based entry can beat random on a non-drifting instrument                                                                                         |
| <code>rrai_momentum_long</code> , <code>froth_high_long</code>                      | Macro overlay | ✗ No edge                                                          | positive net but ~0 <b>excess vs buy-&amp;-hold</b> (just bull drift)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | dropped                                                                                                                                                                                          |
| <code>rrai_euphoria_short</code> + all macro shorts                                 | Macro overlay | ✗ Failed                                                           | –1.5 to –3pp excess (shorting the bull)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | dropped                                                                                                                                                                                          |
| <code>dump_fade_DELETION_short</code>                                               | Equity        | ✗ FAILS at full coverage (was a coverage-bias artifact)            | After profiling <b>all 487 pump-authors (25%→100%)</b> the edge <b>flips negative</b> : gone $\geq 0.20 \rightarrow -19.3\%/10d$ ( $n=87$ ), dominated by <b>squeeze tails</b> (IXHL –625%); PIT <code>young_frac</code> short is <b>–39% to –87%</b> (young pumps rip <i>hardest</i> ). The earlier +14.3% ( $n=36$ @ 25% cov) simply hadn't profiled the squeezers. Win rate still 64% but uncapped stock-short tails kill it.                                                                                                                                                                                                                                                                                                                                                                          | <b>dead as a stock short</b> ; only conceivable via <b>puts</b> (caps the –625% tail) — <b>distribution_short</b> is the better-behaved version (price-rolling-over filter dodges live squeezes) |
| <code>distribution_short</code><br>(concentrated pump + sentiment–price divergence) | Equity        | ✓ <b>The one validated edge — robust across regimes (via puts)</b> | <b>+11.78%/trade naked (win 77%, <math>t=3.57</math>, PF 3.84, Sharpe 2.27). REALISTIC PUTS ~+5%/trade, PF ~2.6</b> after ~12-15% premium drag (these micro-caps have 150-300% IV) — the "+13%/PF 6 puts-capped" figure is an OPTIMISTIC loss-cap model (floors downside without charging premium); plan around +5%. <b>Positive in EVERY regime via puts</b> (2021 +3.5% / 2022 +18.7% / 2025-26 +21%) — the earlier "dead-in-2021" label was ONE uncapped squeeze (BFRI –115%); 2021 median +3%, win 64%. Naked PF robust (excl top-5 winners still 2.66 — the 77% win rate drives it, not 1 trade). Generalizes micro→small→mid, <b>REVERSES on large/mega</b> (–1.9%, win 31%); strongest in liquid \$5-15 band ( $t=5.26$ ). <b>MT5-validated on broker prices</b> (36/37 names tradeable, +11.02%). | forward-OOS clock accruing; <b>trade liquid small/mid names via PUTS, tight-spread only</b>                                                                                                      |
| ~~ <code>frothy_retail_regime</code> ~~<br>(regime filter / kill-switch)            | Meta / regime | ✗ <b>NOT NEEDED — resolved</b>                                     | Investigated for a mania detector; the premise was wrong. The "2021 regime failure" was a <b>single uncapped squeeze (BFRI –115%)</b> dragging the mean to 0 — the 2021 <b>median is +3%, win 64%</b> (signal works fine). Re-squeeze rate is <i>lowest</i> in 2021; IWM-hedge $\approx$ SPY-hedge. <b>PUTS cap the tail</b> $\rightarrow$ 2021 +0.06% $\rightarrow$ <b>+3.46%</b> , other regimes unchanged, overall +13.3%. A gate would wrongly switch OFF a positive 2021 edge.                                                                                                                                                                                                                                                                                                                       | <b>closed</b> — no detector; the always-puts rule already handles the only regime-specific risk                                                                                                  |

| Strategy                                                  | Class         | Status                                             | Headline result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Next action                                                                                                                                             |
|-----------------------------------------------------------|---------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>author_alpha</code> (predictive-user track records) | Equity / meta | ● <b>Persists statistically, NOT tradeable PIT</b> | Split-half persistence is real ( <code>backtest/author_alpha.py</code> : corr +0.27, 8.7% clear $t > 2$ vs 2.5% chance) — BUT the walk-forward PIT fade ( <code>backtest/author_fade.py</code> , 30d maturation buffer, costs) <b>deflates it</b> : FADE-bad $-0.56\% \approx$ FADE-all $-0.60\%$ (author quality adds ~nothing); FOLLOW-good <b>-4.17%</b> (trailing-good is anti-predictive). Works only in <b>2021 mania</b> (+3.13%, $t=16$ ) and <b>loses in current regime</b> ( $-1.10\%$ ). The 9.7pp split-half spread was gross/in-sample. | regime-gate (mania-only fade, single-regime) or extreme-worst-author filter; otherwise shelved — statistical $\neq$ tradeable                           |
| ~~ <code>smallcap_pump_fade</code> ~~                     | Equity        | ✗ <b>NOT an independent edge (gate-ladder)</b>     | The apparent robustness was <b>universe-dependent</b> : on the curated <code>smallcap</code> sector list it looked robust, but on a broad clean universe the same concentration gate is <b>-0.22% and flips negative in 25-26</b> . Concentration alone is $\sim 0$ ; <b>the ROLLOVER gate ( <code>trailing-5d &lt; 0</code> ) is the entire edge</b> (C +2.95% $\rightarrow$ D +12.69%, $t\ 1.0 \rightarrow 4.1$ ). It's just <code>distribution_short</code> with a looser concentration threshold.                                                | folded into <code>distribution_short</code> as a <b>capacity knob</b> (E: $p \geq 3, m \geq 8$ + gates $\rightarrow n=81$ , +9.5%, all-regime-positive) |
| <code>fade_organic_short</code>                           | Equity        | ✗ <b>One-period artifact (revalidated)</b>         | Passes selection null (100th) but regime-split shows the edge is <b>2024-only</b> (+23%); $\sim 0$ every other regime                                                                                                                                                                                                                                                                                                                                                                                                                                | dropped                                                                                                                                                 |
| <code>pump_long</code> (ride)                             | Equity        | ✗ <b>Regime LOSER (revalidated)</b>                | Passes selection null but regime-split: <b>-2/-10/-19/-12% in 2021-24</b> , +14% only 25-26 — momentum-chase that loses when momentum reverts                                                                                                                                                                                                                                                                                                                                                                                                        | dropped                                                                                                                                                 |
| <code>dump_fade_short</code> (naive)                      | Equity        | ✗ Failed                                           | -9.9% (squeezed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | superseded by deletion-gated                                                                                                                            |
| <code>organic_long</code>                                 | Equity        | ✗ Failed                                           | -4.2%, sign-stable loser                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | inverse $\rightarrow$ <code>fade_organic_short</code>                                                                                                   |

## Build & data progress

| Component                                                                         | Status                                      | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature store ( <code>feature_daily</code> / <code>aggregate_daily</code> )       | ✓ Done                                      | ~77k ticker-day rows, point-in-time                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Friction-aware backtester ( <code>backtest/engine.py</code> )                     | ✓ Done                                      | costs + regime split + no-lookahead                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>/signals</code> feed (multi-strategy, json mt5)                             | ✓ Done                                      | strategy-tagged; live, currently <b>flat</b>                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Macro strategy R&D ( <code>backtest/macro_research.py</code> )                    | ✓ Done                                      | only <code>retail_fear</code> beats buy-&-hold; momentum/froth no edge                                                                                                                                                                                                                                                                                                                                                                                                                |
| MT5 framework ( <code>SignalClient.mqh</code> + <code>RedditMacro_EA.mq5</code> ) | ✓ Done                                      | strategy-selectable thin client; demo-only                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Author profiling coverage                                                         | 📊 24% overall / <b>100% of pump-authors</b> | 46,637 profiled, 6,153 gone; sweep now skips crypto-only + front-loads ticker-mentioners. <b>Targeted-profiled all 487 equity pump-authors.</b>                                                                                                                                                                                                                                                                                                                                       |
| Deletion-gated short                                                              | ✗ <b>Failed at full coverage</b>            | The +14.3% was a coverage artifact (25% sample missed squeezers). At 100% pump-author coverage: gone $\geq$ 0.20 – <b>19.3%/10d</b> (IXHL –625% tail); young_frac short –39%→–87%. Stock-short dead; puts-only if at all.                                                                                                                                                                                                                                                             |
| Short-interest / squeeze data (FINRA Reg SHO)                                     | ✓ Done                                      | <code>scraper/finra.py</code> : 475k rows / 5,693 tickers; <b>z-score</b> screen (abs level is MM noise, median 0.49)                                                                                                                                                                                                                                                                                                                                                                 |
| Independent fear-confirmation gate (Wikipedia)                                    | 🟡 → 🟢<br>Strengthened on 2016-26            | <b>Extended series (525 caps, <code>backtest/fear_gate.py</code>): fear-only</b> excess SPY +0.57 (day t=1.64, <b>episode t=2.77</b> ) / QQQ +0.80 (day t=1.91, <b>episode t=2.37</b> ) — effect <i>grew</i> with the 2018 episodes & SPY/QQQ now <b>agree</b> ⇒ likely real (day-level still marginal). The LIVE <b>trend+fear still ≈0</b> (SPY +0.10/QQQ +0.11) → <b>trend filter should be dropped, use fear-only</b> . AUDJPY stays ungated (fear <i>hurts</i> it, +0.21→+0.09). |
| Live screens in <code>/signals</code> ( <code>tradeable:false</code> )            | ✓ Done                                      | <code>smallcap_pump_fade</code> · <code>fresh_pump_alert</code> · <code>coordination_flag</code> · <code>biotech_catalyst</code> · <code>attention_fade</code> · <code>calendar_overlay</code>                                                                                                                                                                                                                                                                                        |
| Idea backlog                                                                      | 💡 7 ideas                                   | squeeze · options-flow · hype-cycle · sentiment-extremes · rotation · novelty · coordination                                                                                                                                                                                                                                                                                                                                                                                          |
| Equity/options paper client (Alpaca)                                              | 💡 Planned                                   | not started                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Methodology (non-negotiables)

1. **No lookahead** — features at date D use only data  $\leq$  D; entries at the first close *after* D.
2. **Frictions** — round-trip spread (small-caps wide), short borrow, halts, capacity.
3. **Regime split** — net return per sub-period; sign-stability is the bar, not a big average.
4. **Shorts via PUTS**, not naked stock — penny shorts have squeeze/borrow/halt tails.
5. **Overlapping-window caveat** — daily signals with N-day forward returns inflate t-stats.
6. **Benchmark macro vs buy-and-hold** — in a bull market "long anything" looks like alpha; a macro long must beat *unconditional drift* (excess), not zero. (Cost us 5 false "survivors".)

## Asset-class reality (what this data can/can't trade)

- **Small/micro-cap & meme equities** — primary signal (retail flow moves price).
- **Single-name options** — primary *expression* (convex longs, defined-risk shorts).
- **Risk-FX (AUD/NZD complex)** — the **primary macro expression**: retail capitulation → contrarian bounce, validated across the cross-section (AUDJPY/NZDJPY/AUDUSD/AUDCHF, t 2.7-3.2). The cleanest tradeable read of the risk-appetite factor RRAI

measures.

- **Index / metals** — *macro overlay* (Gold robust; equities the weak expression,  $t \approx 1$ ).
- **Haven/non-risk FX (USDJPY, CHFJPY, EURUSD) / "Reddit sentiment on SPY as direction"** — no edge; don't build it (they're the controls that confirm the factor).

## 1. Macro overlay — Retail Risk-Appetite Index (RRAI)

Aggregate retail positioning as a **contrarian** factor at extremes. Built posts-only + ratios (coverage-stationary); 90-day trailing percentile. `analytics/features.py::aggregate_daily`.

**Macro R&D verdict ( `backtest/macro_research.py` , **excess over buy-and-hold** ).** A bull market made *every* macro long look like a winner and *every* short a loser — until benchmarked against unconditional drift. Net of that, only the **contrarian retail-fear** family beat buy-and-hold. RRAI-momentum and froth had  $\sim 0$  excess (pure drift); all shorts were  $-1.5$  to  $-3$ pp (fighting the trend). **One real macro edge, three formulations of it.**

### ✓ `retail_fear` — buy risk when retail capitulates (*served live*)

- **Thesis:** retail max-bearish (RRAI pct low / bear-share extreme) + genuine fear (VIX high) → contrarian bounce. Captured 3 ways (capitulation, bear-extreme, pessimism-fade) — all positive-excess, mutually corroborating.
- **Spec (shipped):** long when `rrai_pct`  $\leq 0.15$  **AND** `VIX`  $\geq 18$  ; hold  $\sim 10$ d; size by extremity; **fear-side only**.
- **Evidence (12mo, net, 10d): excess over buy-&-hold** SPY +0.17pp / QQQ +0.23pp / **AUDJPY +0.47pp** (win 88%); `bear_extreme` +0.24/+0.32/+0.36pp corroborates. **VIX gate  $\sim 4\times$ 'd it** (SPY +1.56%/10d in fear vs +0.42% calm). Modest **timing tilt**, not standalone alpha (buy-&-hold already won 66–75%).
- **Instrument scan (17 MT5 instruments, 10d excess vs buy-&-hold):** the signal **generalises across the risk-on basket** — Silver +2.22pp (lumpy, one big rally → size small), Gold +0.59pp (robust, +ve every regime), AUDJPY +0.47pp (steadiest, win 88%), SP500 +0.17pp. Oil (+1.82pp but one-episode) and crypto (BTC net-negative) **dropped**. Generalisation across assets is itself evidence it's a real risk-appetite signal.
- ★ **Cross-sectional FX validation (2026-06-03, effective 2018-26 VIX-gated — the  $VIX \geq 18$  gate has *no* signals in the calm 2016-17, so it starts Feb-2018/Volmageddon) — proves it's a RISK-ON-CURRENCY factor, not an AUDJPY fluke.** Ran capitulation-long across the yen-cross + risk-pair complex. The edge travels with the **risk-on currency (AUD/NZD)**, *independent of the funding leg* — and the **controls fail**, which is the real proof.

| pair         | character            | excess | t(exc)              |
|--------------|----------------------|--------|---------------------|
| NZDJPY       | risk-on JPY cross    | +0.43% | <b>+3.18</b>        |
| AUDJPY       | risk-on JPY cross    | +0.48% | <b>+2.93</b>        |
| AUDCHF       | risk vs CHF-haven    | +0.41% | <b>+2.83</b>        |
| AUDUSD       | risk pair (no JPY)   | +0.43% | <b>+2.72</b>        |
| EURJPY       | risk-ish             | +0.25% | +2.70               |
| NZDUSD       | risk pair            | +0.35% | +2.33               |
| Nikkei ^N225 | high-beta JP equity  | +0.62% | +1.93               |
| USDJPY       | JPY cross, USD~haven | +0.12% | <b>+1.02</b> (flat) |
| CHFJPY       | CONTROL haven/haven  | +0.13% | <b>+1.37</b> (flat) |
| EURUSD       | CONTROL non-risk     | +0.14% | <b>+1.44</b> (flat) |

**Read:** every AUD/NZD pair is significant ( $t$  2.7–3.2) regardless of funding leg; the three pairs with *no* risk-on leg (USDJPY/CHFJPY/EURUSD) are the three weakest. **It's not a JPY thing** — JPY is just a clean funding leg; the Nikkei works because it's high-beta *risk*, not because it's Japanese. Equities stay the weak expression (SPY  $t=0.90$ , QQQ  $t=1.13$ ). The clean cross-section (right

pairs fire, controls don't) is strong evidence of a genuine factor → **trade the basket, not one pair.** - 🟡 **Independent fear-confirmation gate (Wikipedia, LIVE — but now DOUBTFUL):** the equity legs (US500/USTEC) additionally require an *independent* fear spike —  $\text{fear\_z} \geq 0.5$ , the z-score of Wikipedia fear-page attention (Stock\_market\_crash / Recession / Bear\_market views) vs the trailing ~30d — **AND** a 200-DMA uptrend. Original rationale: SPY capitulation *alone* doesn't beat buy-and-hold (master backtest excess -0.05); an *ad-hoc* run reported **+0.36pp/10d when fear\_z confirms vs ~-0.01 ungated**, and fear-attention is **1.32x higher on capitulation days** ( DATA\_PLAN.md ). Wired live in analytics/signals.py ( \_fear\_z , FEAR\_MIN ; fear\_z in /signals ). - 📄 **MT5 Strategy-Tester A/B (US500, D1, 2020–2026, long-only, real spread/swap; mq15/RedditMacro\_US500\_\*.ini , dates via python main.py --export-signals ).** This tests the gate as an **absolute long-only rule** (NOT excess-vs-buy-&-hold):

```
| variant | signals | trades | win% | net $ | PF | maxDD% |
|-----|-----|-----|-----|-----|-----|-----|
| baseline (all caps) | 335 | 100 | 57.0 | **+12** | 1.00 | 13.6 |
| 'fear_z≥0.5' only | 95 | 45 | 53.3 | **~1045** | 0.68 | 13.0 |
| trend up (200-DMA) only | 223 | 71 | 56.3 | **~21** | 1.01 | 9.0 |
| trend **AND** fear (the live gate) | 54 | 27 | 55.6 | **~372** | 0.73 | 7.1 |

**Result: the 'fear_z' gate does NOT help - it hurts.** Adding fear makes US500 *worse* both alone (-$1045)
```

- ☒ **Committed EXCESS study ( backtest/fear\_gate.py , python -m backtest.fear\_gate ) — this settles it.** The frame MT5 can't measure: mean EXCESS over buy-&-hold (the +0.36pp's own metric). Same point-in-time date-sets as the MT5 run (shared capitulation\_sets() ). 10d, net of costs:

| leg                       | variant                    | n   | excess%      | win% | t(exc)       |
|---------------------------|----------------------------|-----|--------------|------|--------------|
| SPY                       | baseline                   | 332 | -0.05        | 61   | -0.21        |
| SPY                       | <b>fear</b>                | 94  | <b>+0.32</b> | 68   | +0.64        |
| SPY                       | trend                      | 220 | -0.03        | 60   | -0.16        |
| SPY                       | trend+fear ( <i>live</i> ) | 53  | +0.09        | 64   | +0.19        |
| QQQ                       | baseline                   | 332 | +0.06        | 62   | +0.20        |
| QQQ                       | <b>fear</b>                | 94  | <b>+0.41</b> | 65   | +0.71        |
| QQQ                       | trend+fear ( <i>live</i> ) | 53  | <b>-0.20</b> | 60   | -0.33        |
| AUDJPY ( <i>ungated</i> ) | baseline                   | 333 | <b>+0.26</b> | 68   | <b>+2.04</b> |

**Verdict (three honest parts):** (1) **The +0.36pp REPRODUCES as a point estimate** — fear-only SPY +0.32 / QQQ +0.41 excess — so it was *not* fabricated; the fear gate does lift equity-capitulation excess. (2) **But it is NOT statistically significant** (excess  $t \approx 0.6$ ,  $n \approx 94$ ) — a thin, unproven edge. (3) **The LIVE trend+fear combo is the wrong way to apply it:** the trend filter *removes the buy-fear-in-a-downtrend bounces that carry the excess*, diluting SPY (+0.32 → +0.09) and flipping QQQ negative (+0.41 → **-0.20**). - **Reconciliation (why MT5 said "loss" and this says "small win"):** MT5 sums *absolute* P&L and holds one position at a time, so a few COVID-cluster trades dominate the dollar total (-\$1045); the excess frame means-averages independent trades, where fear is mildly +ve. Both are correct for their question — for *signal research* the excess frame is the right one, so the earlier "fear-gate is a drag" was a frame artifact. - **The genuine tension:** fear → better *timing/excess* but worse *drawdown* (COVID knives); trend → better *drawdown* but worse *excess*. The live **trend AND fear** gets the worst of both for excess. - **Recommendation:** keep the gate 🟡 (positive but insignificant). If we ever optimise the equity legs for excess, use **fear-only, not trend+fear**; AUDJPY (the one significant edge, +0.26  $t=2.04$ ) stays **ungated** — confirmed correct. (User chose to leave the live gate unchanged for now; this is logged for when we revisit.) - 📄 **Power / effective sample (day vs episode — the divergence IS the verdict):** the 94 fear-days are *not* 94 independent obs — overlapping 10d windows + episode clusters. Treated as **days**, fear-only is  $t=0.64$  (SPY)/0.71 (QQQ); collapsed to **independent episodes** (~40, one vote each) it's  **$t=2.28$  (SPY)/1.85 (QQQ)**. The gap is driven almost entirely by the **COVID-2020 cluster** (~10+ losing fear-days → one episode vote), so the effect is *positive across most episodes but fragile* — and SPY (2.28) vs QQQ (1.85) disagreeing on near-identical instruments over the same episodes says we're at the **noise floor**, not a stable edge. **Data to settle it:** day-level needs ~9x **more** obs for  $p < 0.05$  (~19x for 80% power) — impractical (~40+ yrs at ~15-20 fear-days/yr); the real lever is **independent fear**

**episodes**, and the cheapest source is *backward* (Wikipedia fear data already reaches 2015) — **extending the RRAI/capitulation series to 2016-2017 adds the 2018 Volmageddon + Q4-2018 episodes** (price coverage starts 2016-05). -  **DONE — RRAI extended to 2016-06 (525 caps, +190)**. Re-ran the study on the longer series, and the fear-gate **strengthened** (effect *grew* out-of-sample — the hallmark of a real edge):

| leg | fear excess (94d → 141d) | day t (→)   | episode t (→)      |
|-----|--------------------------|-------------|--------------------|
| SPY | +0.32 → <b>+0.57</b>     | 0.64 → 1.64 | 2.28 → <b>2.77</b> |
| QQQ | +0.41 → <b>+0.80</b>     | 0.71 → 1.91 | 1.85 → <b>2.37</b> |

**Both equity legs are now episode-level significant (t≈2.4-2.8) AND agree** — the earlier SPY≠QQQ noise-floor red flag is gone. Day-level is still only marginal (1.6-1.9), so not fully nailed, but this is a real upgrade from "fragile/unproven." The trend filter conclusion **hardens**: `trend+fear` (live) is still ≈0 excess (SPY +0.10/QQQ +0.11) ⇒ **drop trend , use fear-only** on the equity legs. AUDJPY confirmed best ungated (baseline +0.21 t=2.00; fear-gating *hurts* → +0.09). - **Expression (GOLD-LED basket, finalised 2026-06-03 after MT5+OOS): Gold (XAUUSD) primary + AUDJPY small secondary**. The excess t-stats liked the whole AUD/NZD complex, but realistic execution + OOS kept only these two: **AUDUSD dropped** (failed OOS, PF 0.47), **NZDJPY dropped** (no OOS), US500/USTEC/Silver dropped earlier (no alpha). `euphoria_short` parked. Served via `/signals` (tag `retail_fear`); traded by `mqL5/RedditMacro_EA.mq5` with **per-instrument params** — Gold StopATR 3 / hold 11 / risk 1.0%; AUDJPY StopATR 2 / hold 24 / risk 0.5% (its OOS-optimal long hold, half size as a secondary). - **Caveats**: single ~12mo bull regime; **46 cap-days = only ~13 distinct fear episodes** (clustered → effective n small); overlapping windows; edge small in absolute terms. - **Next**: leverage-language enrichment (calls/puts, margin/YOLO); longer history (data-limited); vol-target sizing; combine capitulation+bear-extreme into one composite trigger.

 **Macro candidates that failed (kept so we don't retry)**

- `rrai_euphoria_short` , `bull_extreme_short` , `froth_high_short` , all momentum-shorts: **-1.5 to -3pp excess** — shorting a bull market.
- `rrai_up_momentum_long` , `froth_high_long` : **~0 excess** — just drift, no signal.

## 2. Equity cross-sectional (per ticker × day)

From `analytics/features.py::feature_daily` (velocity, accel, breadth, concentration/HHI, sentiment trajectory, author young/gone/suspended mix, novelty).

 **`pump_long` — ride concentrated promotion**

- **Thesis**: concentrated mention spike (few authors, high volume) pops short-term.
- **Result**: net +3%/5d but **sign-flips** (P1 +15.8 → P3 -4.6), win 36%, t=0.75. The crude "+6.45%" was a no-lookahead/no-cost/single-regime artifact. **Not robust**.

 **`dump_fade_short` — short a decelerating pump (naive)**

- **Result**: net **-9.9%** (P1 -29%). Shorting pumps on deceleration gets squeezed/run over. Wrong trigger; brutal without defined risk.

 **`organic_long` — buy broad attention spikes**

- **Result**: sign-stable **-4.2%** net (t=-4.25, n=321). Buying broad/news-driven attention tops is a *consistent loser* → the inverse is the candidate (below).

 **`fade_organic_short` — fade broad attention spikes**

- **Thesis**: broad attention spikes mean-revert down (inverse of the robust `organic_long` loss).
- **Result**: net +0.83%, but **flips in P3** (win 46%) — the reversion is real but shorting costs + recent weakening eat it. **Marginal**. Revisit via **puts** (cheaper than borrow) and with a VIX/vol gate.



✗ `dump_fade_DELETION_short` — short a pump whose promoters vanished (*THE thesis — DIED at full coverage*)

⚠ **REVERSAL (2026-06-03)**. This was "forensically validated" at +14.3%/10d (n=30-36) — but that ran at only ~25% pump-author coverage. We then **targeted-profiled all 487 equity pump-authors (→100%)** and re-tested: the edge **flips to -19.3%/10d (n=87)**, and the PIT `young_frac` short is -39% to -87%. Cause: **squeeze tails** — a single ticker (IXHL) squeezed **-625%/-404%/-377%** in July-2025; the partial sample simply hadn't profiled its authors, so those events were invisible. Lesson logged: **small-n + partial-coverage results are dangerous**; shorting live manipulated micro-caps has *uncapped* left-tail risk. Win rate is still ~64% (most pumps fade), so the thesis isn't *wrong* — but it's only expressible via **puts** (defined risk caps the -625%). The price-rolling-over filter is what makes `distribution_short` survive where this dies. - **Thesis (original)**: concentrated pump + promoters now **deleted/suspended** → manufactured → fade the collapse. The original A2 idea, and the *reason* we track account lifecycle. - **Result (2026-06-02, 20% author coverage, h=10, market-neutral, 75bps + 8bps/day borrow)**: | gate | n | net short | win | t | regimes | |---|---|---|---|---|---| | `gone_frac`≥0.10 | 33 | +10.0% | 81.8% | 1.51 | -1.2 / +19.9 / +11.4 | | `gone_frac`≥0.20 | 30 | +14.3% | 83.3% | 2.57 | +11.4 / +27.7 / +3.9 | | **control**: pumps w/ promoters *still here* ( `gone`=0 ) | 202 | -10.8% | 36.1% | -2.22 | -30.7 / -0.6 / -1.1 | - The gate **flips the sign of the trade** (~+25pp spread). Un-gated concentrated pumps are *un-shortable* (they continue / squeeze); only the ones whose promoters vanish collapse. The deletion is doing the work, not pump-selection. Positive across all 3 regimes at `gone`≥0.20. - ⚠ **Not yet point-in-time tradeable**. `gone_frac` uses *hindsight*: Reddit exposes no deletion timestamp, so "gone" means "gone as of our latest sweep" — unknowable at the pump date. The PIT-available leading proxy is `young_frac` (account age at post time), but it's currently too sparse to conclude (n=3-7, and the few events lean the wrong way). - **Path to tradeable**: (1) **finish the author sweep** (20%→40%+) to power the `young_frac` PIT test; (2) failing that, a **live deletion-monitor** rule — `author_status_revalidate` already detects deletions forward; short on deletion-*confirmation* within K days of a fresh pump, capturing the back half of the collapse. Express via **puts** (defined risk vs squeeze tails; many are HTB/no-borrow).

● `distribution_short` — concentrated pump + sentiment-price divergence (*NEW, point-in-time tradeable*)

- **Thesis**: the "bagholder distribution" moment — a *manufactured* pump (concentrated chatter) where **price has started rolling over but the crowd is still euphoric**. Insiders distribute into the retail bid; sentiment lags price. Every input ( `per_author` , `mentions` , `sentiment` , trailing price return) is known at signal time → **unlike the deletion gate, this is actually tradeable live**. ( `backtest/microstructure.py` )
- **Rule**: `per_author`≥2 AND `mentions`≥10 AND `sentiment`≥0.4 AND `5d-trailing-return`<0 → SHORT. h=10, market-neutral, 75bps + 8bps/day borrow.
- **Result (2026-06-02)**: +7.2%/10d net, t=2.70, win 52%, n=60 across 39 distinct tickers.
- **Date-clustered test** (1 obs/date, kills intra-day correlation): +7.6%, t=2.52, 47 independent dates — *not* a risk-off-day artifact (max 3 events on any date, spread over 12 months).
- **Dose-response** in sentiment (s≥0.3/0.4/0.5 → +3.9/+7.2/+11.1%) and **monotone in horizon** (h=5/10/20 → +5.6/+7.2/+9.5%, t rising to 2.92) — the hallmarks of a real effect, not a fit.
- **Survives 2x costs** (150bps spread + 16bps/day borrow → still +4.9%/10d).
- **Control**: same concentrated pumps *without* the divergence (price up) → -4.1% — the divergence is doing the work. Positive in all 3 sub-periods.
- **Why it works where the others don't**: the gate is *precise* — it only fires inside the manufactured subset. The broad high-attention universe **drifts flat-to-up** (gross ≈ +0.04%), so generic pump-shorting just pays costs (see negatives below); the edge requires isolating the distributing pumps, which both this and the deletion gate do.
- **Caveats / next**: n still modest (47 indep. dates); single ~12-mo window (sub-periods, not true bull/bear regimes); small-cap-heavy so no multi-regime price history; loose `per_author`≥2 lets some ETFs/large-caps (QQQ/IWM) leak in — tighten to ≥3 and/or split small-cap vs index. **Collect forward OOS** (the clock starts now); express via **puts**. Not yet promoted to the live `/signals` feed.

✗ **Negatives from the same study (recorded so we don't re-run them)**

- **Broad-universe microstructure timing does NOT survive costs**. Recruitment-exhaustion, mention-deceleration, and sentiment-price divergence applied to the *whole* high-attention set (`mentions`\_z≥1) are all **net-negative** — the universe drifts



flat-to-up, so shorting it generically just bleeds the ~2.3% friction. Precision (concentration filter) is mandatory.

- **Recruitment-exhaustion alone = null.** New-author inflow rolling over is directionally supportive *within* concentrated pumps (+1.8% net, but  $t=0.33$ ) and adds to the stack, but is not a standalone edge at current coverage. The new-author feature is built and kept.
- **Up-leg longs both fail.** (a) "Ride while recruiting + accelerating" (concentrated) netted +1.6% but win 27%, regimes [+21/-7/-9] — one big winner, otherwise negative. (b) "Attention-leads-price coil" (broad: recruitment surging while price still flat) was **net -1.4 to -1.9%** across horizons, sign-stable loser. Neither is a tradeable long; dropped.

### 3. Idea backlog (💡 not yet built)

- **Squeeze setup (long):** Reddit squeeze chatter (Shortsqueeze/SqueezePlays) + **short pressure** → squeezable. *Partially built:* `scraper/finra.py` + `squeeze_screen()` flag tickers with an anomalous **z-score jump** in daily short-ratio vs their own baseline (HTZ/GME currently surface). *Still needs* true SI%/float/days-to-cover (FINRA **bi-monthly short-interest** report) + borrow-fee for the level; the daily file's absolute level is market-maker noise. *Not yet backtested.*
- **Options-flow signal:** parse contract-level chatter (unusual\_whales/options) → retail-gamma/positioning proxy. *Needs contract parsing.*
- **Hype-cycle lifecycle:** model attention as nascent→acceleration→peak→fade; long the *organic* acceleration, fade the *concentrated* peak. Breadth-vs-concentration is the discriminator.
- **Conditioned sentiment extremes (single-name):** linear sentiment is dead, but *extreme concentrated euphoria at a price high* (short) / *capitulation after a sharp drop* (bounce) may be non-linear edges.
- **Cross-sectional attention rotation:** rank meme universe by abnormal velocity z-score; long top-decile / short bottom-decile, beta-hedged with SPY; baskets from co-mention clusters.
- **Novelty / first-mention alpha:** the "discovery" moment (silent → sudden organic burst).
- **Coordination graph (short):** rigorous version of the deletion short — cluster density + temporal bursts + duplicate text + new-account% + gone%.

### 4. Data gaps blocking strategies

| Gap                                           | Unblocks                                                                                                                                                      |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Intraday prices</b> (have EOD only)        | timing pump/dump entries/exits (they move intraday)                                                                                                           |
| <b>Short interest / float / borrow fee</b>    | squeeze <i>level</i> (FINRA daily short-volume now in via <code>scraper/finra.py</code> , but it's MM noise at the level; need bi-monthly SI%/float + borrow) |
| <b>Options chains / IV</b>                    | options expression; options-flow signal                                                                                                                       |
| <b>Longer history</b> (data-limited to ~12mo) | true multi-regime validation of the RRAI overlay                                                                                                              |
| <b>Author coverage</b> (20% → target ~40%+)   | a <i>point-in-time tradeable</i> deletion short — thesis already forensically validated; PIT <code>young_frac</code> proxy needs more coverage                |

### 5. Execution stack

- `/signals feed` ( `analytics/signals.py`, GET `/signals json|mt5` ) — live, serves the capitulation overlay. ✅
- **MT5 EA** ( `mq15/RedditRRAI_EA.mq5` ) — thin client for the index/FX overlay; demo-only. ✅
- **Alpaca / IBKR paper client** — for the equity/options legs. 💡 planned.

## ★ MULTI-REGIME VALIDATION (2020-2026, incl. COVID crash + 2022 bear)

The deep-history backfill let us test across regimes — the result that actually moves confidence: - **AUDJPY capitulation-long is REGIME-ROBUST** — positive in EVERY year 2020-2026, including the 2022 bear (+0.7%, win 60%), win rates 60-93%. Graduates from preliminary → **validated**. This is the core tradeable edge. - **Equity (SPY) capitulation-long FAILS in a sustained bear** (2022: -0.7%/10d, win 43%) — falling-knife problem → **must gate with the 200-DMA trend filter** (don't buy fear in a downtrend). Confirmed needed. - **Euphoria-short only works in the bear** (2022 +1.3%; loses every bull year) → shorts are **regime-conditional**, switch on only in confirmed downtrends. (Vindicates not building shorts earlier.) - Caveats: small n/year (18-60), overlapping windows, historical RRAI is a WSB-heavy sampled backfill.

## Session findings — sector / calendar / latency (2026-06-01)

- **Sector segmentation rescues the pump signal** ( `backtest/sectors.py` ). The broad pump was a wash because it *mixed opposites*: **small-cap concentrated pumps FADE** (−14.6%/10d excess, win 22%,  $t = -3.93$  — pump-and-dumps reliably reverse → a short/avoid), while **biotech pumps are a fat-tailed lottery** (+66% mean, win 48%,  $t = 2.23$  — a few catalyst moonshots). Thematic universes (ai/quantum/space/nuclear/meme/frontier) had too few concentrated-pump events to judge.
- **Pump detection is TOO SLOW** ( `feature_daily` ). Median `novelty_days` at flag = **152**; only **1%** of pumps flagged within 3 days. Root cause: the `min_total ≥ 20` universe filter + trailing z-windows mean a ticker must accumulate ~20 mentions before it's even eligible → we catch *established-ticker spikes*, not fresh pump-and-dumps. Only **33 fresh day-0 bursts** exist in the data. **Fix**: a fresh-burst fast-path that scans raw mentions (ticker's first day with a burst), bypassing the 20-mention gate — for LIVE catching, not just post-hoc.
- **Calendar effects (10y, SPY/QQQ/Gold; absent in AUDJPY-FX)**: turn-of-month is real (SPY +0.49 vs +0.20%/5d, QQQ +0.58 vs +0.35, Gold +0.38 vs +0.25); September weak (−0.2%); OpEx week underperforms. Generic equity-flow seasonality → useful as *overlays/filters*, not Reddit alpha.
- **AU sentiment track** building ( `au_aggregate_daily` , post-level) to test AU-sentiment → AUDJPY.

## ★ MASTER BACKTEST vs BUY-AND-HOLD ( `backtest/run_all.py` , full 2020-2026, 10d, net of costs)

Every signal judged on EXCESS over its instrument's unconditional drift (buy-and-hold-anytime). | Strategy | n | net% | B&H% | excess | win | verdict | |---|---|---|---|---|---| | retail\_fear **AUDJPY** | 338 | +0.53 | +0.27 | **+0.25** | 69% | beats (modest, most robust) | | retail\_fear SPY | 338 | +0.54 | +0.59 | −0.05 | 61% | **no edge** | | retail\_fear Gold | 338 | +0.83 | +0.72 | +0.10 | 57% | marginal | | euphoria\_short SPY | 351 | −1.26 | +0.59 | −1.84 | 27% | no edge | | smallcap\_pump\_fade | 87 | +9.07 | +0.59 | +8.48 | 64% | beats but single-regime + brutal costs | | biotech\_catalyst | 91 | +22.2 | +0.59 | +21.6 | 36% | fat-tailed LOTTERY (low win) | | attention\_fade | 252 | −0.73 | +0.59 | −1.31 | 40% | no edge |

**Recalibration**: with a proper buy-and-hold benchmark over the full period, the AUDJPY edge is a **modest +0.25pp/10d** (earlier raw "positive every year" overstated it - that was largely drift). SPY capitulation does NOT beat buy-and-hold. The big-excess equity signals are single-regime (smallcap) or low-win lotteries (biotech). Net: one modest-but-real macro edge, several promising- but-unvalidated screens. Do NOT size up on these as-is.

## DATA WE NEED (to validate/strengthen — the binding constraints)

| Data                                                                               | Unblocks                                                                                                                           | Why we lack it                                                                                                           |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Intraday prices</b> (we have EOD only)                                          | pump entry/exit timing; fresh-pump fade execution (pumps move intraday)                                                            | not collected                                                                                                            |
| <b>Historical penny-stock prices + per-ticker features</b>                         | multi-regime validation of smallcap_pump_fade & biotech (only 2025-26 exists)                                                      | delisted pennies → Yahoo gaps; per-ticker feature_daily only recent                                                      |
| <b>Short interest / float / borrow fee</b>                                         | squeeze setups; realistic short cost/feasibility                                                                                   | FINRA daily short-volume in ( <code>scraper/finra.py</code> ); still need SI%/float/borrow (Ortex / FINRA bi-monthly SI) |
| <b>Options chains / IV</b>                                                         | express fades via PUTS (only sane way to short pennies); options-flow signal                                                       | not collected                                                                                                            |
| <b>More years of RRAI history</b>                                                  | tighten the macro edge (effective $n \approx$ a few dozen fear episodes)                                                           | Reddit/arctic-shift depth ~2017+                                                                                         |
| <b>Broader author coverage</b> (~20% profiled)                                     | a <i>PIT-tradeable</i> deletion/young-account short (thesis already validated forensically; <code>young_frac</code> proxy starved) | rate-limited Reddit /about sweep                                                                                         |
| <b>Independent sentiment sources</b> (StockTwits, news, Discord, crypto + AU subs) | independent episodes → real effective-sample growth without waiting calendar time                                                  | not integrated (AU track building)                                                                                       |
| <b>Real transaction-cost data</b> (per-instrument spreads/borrow)                  | trustworthy net returns (small-cap costs are modelled/optimistic)                                                                  | not sourced                                                                                                              |

## Update — complete-data confirmation (2026-06-01)

- US backfill finished: continuous 2020-2026 (2,344 days, 341 capitulation days). **Capitulation-long multi-regime result HOLDS on the complete series** — AUDJPY positive every year incl. 2022 bear (win 60-83%).
- `smallcap_pump_fade` is strong (+12.4% excess short, win 77%) but **only 2025-26 data exists** (n=87) — no historical penny-stock prices/features, so it **can't be multi-regime validated** → stays ADVISORY, NOT promoted to tradeable. (Single-regime; can't be fixed by backfill — penny price history doesn't exist.)

## Crypto — coincident, NOT a leading indicator (tested 2026-06-03)

Added Cryptocurrency/Bitcoin/CryptoMoonShots/SatoshiStreetBets (~1yr; Bitcoin 61k posts) and tested whether crypto leads equities / risk-appetite. **It doesn't — it's coincident**: - **Price lead-lag (BTC↔SPY, 2,577d)**: same-day corr **+0.17 (t=8.8)** but every lead/lag  $k \neq 0$  is noise ( $|t| < 1.3$ ) — neither leads. - **Weekend test** (crypto's best shot — BTC trades Sat/Sun while equities are closed; 490 weekends): corr +0.05, **t=1.2** — no lead. - **Crypto sentiment → equity RRAI**: corr **peaks at k=0 (+0.20, t=3.9)** then decays (a lead would peak at  $k > 0$ ). Barely predicts even BTC's *own* next-day return (+0.05). - **Combined crypto+equity RRAI capitulation (1yr overlap, n=55)**: does **NOT** help — for the validated AUDJPY edge it *dilutes* it (equity-only excess **+0.55%** > combined +0.45% > crypto-only -0.03%). Crypto sentiment is redundant with (and noisier than) the equity RRAI. - One weak whiff: crypto *euphoria* → mildly negative SPY 3d (t=-2.4, but overlapping windows → true  $t \approx -1.4$ ) — *contrarian*, not leading. Watch, not an edge.

**Takeaway**: crypto's value is NOT macro timing (it's a redundant coincident risk-appetite read). It's worth keeping only as its own **pump/meme universe** (CryptoMoonShots/SatoshiStreetBets) for the `distribution_short` /pump signals. Do not fold it into the RRAI.

## ★ Cross-asset generalization map (2026-06-03) — it's a RISK-APPETITE / FEAR-REVERSAL factor

Capitulation-long ( $r_{rai} \leq 0.15$  &  $VIX \geq 18$ , 10d, excess vs buy-and-hold) tested across asset classes. *Effective window 2018-02 → 2026-04 — the  $VIX \geq 18$  gate has no signals in the low-vol 2016-17.* The signal fires wherever the instrument is a clean **risk-on** or **fear-bid** play, and is flat everywhere else — the pattern of winners *and* losers is mechanistically coherent (= a real factor):

| asset class                      | works?             | evidence                                                                                             |
|----------------------------------|--------------------|------------------------------------------------------------------------------------------------------|
| <b>Risk-on FX</b> (AUD/NZD)      | ✅ <b>strongest</b> | AUDJPY $t=2.93$ , NZDJPY $t=3.18$ , AUDCHF 2.83, AUDUSD 2.72, NZDCHF 2.58, ZARJPY 2.32 (EM, noisier) |
| <b>Safe-haven metals</b>         | ✅ gold             | Gold +0.43% $t=2.03$ ; Silver +0.76% $t=1.72$ (lumpy)                                                |
| Equity indices (US/AUD/JP)       | ❌ weak             | Nikkei +0.62 $t=1.93$ (best); S&P $t=0.90$ , Nasdaq 1.13; <b>ASX 200 <math>t=0.08</math> (flat)</b>  |
| Growth commodities               | ❌ none             | Copper $t=0.68$ , Platinum 0.87, <b>Oil <math>-0.10</math></b>                                       |
| <b>Controls</b> (should be flat) | ✅ flat             | <b>AUDNZD (risk/risk) <math>t=1.11</math></b> ; USDJPY 1.02, CHFJPY 1.37, EURUSD 1.44                |

**Punchlines:** (1) the AUD currency carries the edge ( $t=2.9$ ) but the AUD equity index (ASX) is flat ( $t=0.08$ ) — it's an FX/metals phenomenon, not equities. (2) **AUDNZD flat** is the cleanest proof: two risk-on currencies cancel the factor. (3) It's ONE factor, so more instruments = diversification *within* it (diminishing), not new alpha; EM/growth-commodity legs add cost/tail-risk for little gain. **MT5 absolute check** (long-only, ungated, BaseLots=1.0): AUDJPY net +\$10.4k (PF 1.14) but AUDUSD ~flat in dollars (it *declined*, so +excess  $\neq$  +absolute) and 60%+ DD → deployable version needs the VIX gate + vol-target sizing.

## ⚠️ MT5 EXECUTION REALITY CHECK (2026-06-03) — the excess edge does NOT survive naive execution

After fixing the EA's position sizing (ATR fixed-fractional, risk 1%/trade) and hold (trading days), the MT5 tester is finally trustworthy (FX-leg drawdowns fell from ~90% to ~17-20%). It tells a sobering story:

| leg (VIX-gated entry, ATR-sized, 10 trading-day hold, 2018-26) | trades | win% | PF          | maxDD%                                |
|----------------------------------------------------------------|--------|------|-------------|---------------------------------------|
| AUDJPY                                                         | 67     | 41.8 | <b>0.85</b> | 20                                    |
| AUDUSD                                                         | 67     | 55.2 | <b>0.91</b> | 17                                    |
| Gold                                                           | 67     | 56.7 | <b>1.01</b> | 93 ⚠️ (no stop → trending losses run) |

**The validated EXCESS edge (AUDJPY  $t=2.93$ , etc.) does NOT translate into a profitable naive long-on-trigger MT5 strategy.** Why the gap: 1. **Entry timing.** Python averages over *all* capitulation days (overlapping); the EA enters on the *first* trigger of a cluster — i.e. while the market is *still falling* (the cluster-start = falling knife). The bounce the excess metric sees is *averaged over the cluster*, not capturable by entering at its start. 2. **Excess  $\neq$  absolute.** The FX legs have ~0 unconditional drift, so a positive excess (+0.48% vs B&H) nets ~0 in *dollars*. The Python t-stat measures "higher than drift," not "profitable." 3. **No stop.** Holding 10 days with no stop lets losses run (gold DD 93%).

**Honest status (before the fix):** the edge is a real *statistical* property but was **not deployable** as a naive long-on-trigger trade. Predicted fix: catch the *turn*, not the first fall, + a stop.

## ✅ RESOLVED — turn-based entry + stop makes it tradeable (2026-06-03)

Added to the EA: **arm on a capitulation signal, then enter on the first up-bar (the turn) within ArmWindow, with a StopATR×ATR stop**; plus selectable sizing (Fixed / ATR / Kelly). MT5 results (VIX-gated → 2018-26):

| symbol | sizing | entry       | win% | PF          | DD%  | Sharpe      |
|--------|--------|-------------|------|-------------|------|-------------|
| AUDJPY | ATR    | naive       | 42.5 | <b>0.77</b> | 9.6  | -0.44       |
| AUDJPY | ATR    | <b>turn</b> | 51.4 | <b>1.06</b> | 7.0  | 0.10        |
| Gold   | ATR    | naive       | 53.5 | 1.25        | 39.7 | 0.29        |
| Gold   | Fixed  | <b>turn</b> | 55.1 | <b>1.54</b> | 33.3 | <b>0.41</b> |
| Gold   | ATR    | turn        | 55.1 | 1.39        | 44.9 | 0.40        |

- **The turn entry works** — it flips AUDJPY from PF 0.77 (losing, naive) to **PF 1.06** (win 42.5→51.4%), and lifts Gold to **PF 1.54 / Sharpe 0.41**. Waiting for the bounce instead of catching the knife is the difference between losing and winning.
- **Gold is the strongest deployable leg** (PF 1.4-1.5, Sharpe 0.4). AUDJPY is now marginally positive (PF ~1.06) — thin but no longer a loser.
- **Sizing verdict: ATR/Fixed win; Kelly OVERSIZES**. Even half-Kelly hit its cap on these thin edges and blew drawdowns to 30% (AUDJPY) / 80% (Gold) with PF≈1.0 — no return benefit. The textbook result: **Kelly is too aggressive for thin/uncertain edges; fixed-fractional ATR (~1%/trade) is the right default**.
- **Still preliminary**: PFs are modest, gold DD is high (33-45%), and this is in-sample on the same series the signal was built on. Forward OOS + a small live demo allocation are the next validation steps. But the basket is now **tradeable, not just a statistical curiosity**.

## MT5 parameter optimization (2026-06-03) — in-sample optimize → OOS validate

Ran the **MT5 genetic optimizer** (headless; every pass logged via frames to `rrai_opt_log.csv`; `mql5/RedditMacro_Optimize.ini`) on Gold, sweeping ArmWindow / StopATR / MaxHoldDays, **in-sample 2018-2022**, maximising net profit — then **validated the winners out-of-sample (2023-2026)**:

| param set                         | IS PF / DD | OOS PF / DD / win       |
|-----------------------------------|------------|-------------------------|
| profit-max (stop 1.0, hold 5)     | 1.52 / 55% | 1.52 / 42% / 52%        |
| <b>robust (stop 3.0, hold 11)</b> | 1.53 / 33% | <b>1.96 / 18% / 62%</b> |
| default (stop 2.0, hold 10)       | —          | 1.75 / 26% / 56%        |

- **Encouraging**: the signal **didn't badly overfit** — every parameter set stayed profitable OOS (PF 1.5-2.0). The gold leg is robust to parameter choice.
- **But "max profit" is the wrong objective**: it picked a tight-stop / quick-exit corner with **2-3× the drawdown** (42-55%) for the same PF. **The robust set (wide stop 3×ATR, hold ~11d) wins OOS: PF 1.96, DD 18%, win 62%** — optimise for *risk-adjusted* return (Sharpe/PF-with-DD), not raw profit. (NB: *this 1.96 used ArmWindow=7; the shipped default arm=5 gives ≈1.27 on the same 16 OOS trades — the figure is arm-sensitive on a tiny sample, see the "MT5 deployable-config report" correction.*)
- **ArmWindow ≥3 is irrelevant** (the turn arrives within ~3 bars). The live params worth using on gold: **StopATR≈3, MaxHoldDays≈11**.

## Per-instrument optimization (AUDJPY + AUDUSD, same IS→OOS protocol) — the deployable reality

Optimised AUDJPY and AUDUSD the same way (genetic IS 2018-22 → OOS validate 2023-26):

| leg         | best-robust (IS)       | OOS PF / net / DD           | verdict                                                                                       |
|-------------|------------------------|-----------------------------|-----------------------------------------------------------------------------------------------|
| <b>Gold</b> | stop 3, hold 11        | <b>1.96 / +\$3.5k / 18%</b> | ✅ <b>robust — the deployable leg</b>                                                          |
| AUDJPY      | stop 2, hold <b>24</b> | 1.31 / +\$171 / 5%          | ⚠️ marginal — only +ve OOS with a <i>long</i> hold; default (hold 10) is <b>-ve</b> (PF 0.87) |
| AUDUSD      | stop 2, hold 5         | <b>0.47 / -\$603</b>        | ❌ <b>overfit — fails OOS</b> (IS PF 1.78 → OOS 0.47)                                          |

- **Only Gold robustly survives** realistic execution + OOS. **AUDUSD fails OOS** (its positive excess t-stat doesn't translate — excess≠tradeable, again — AUDUSD declined 2023-26 so long-only loses in dollars). **AUDJPY is thin/finicky** (needs hold≈24; the shared-basket default hold underperforms it).

- **Implication for the live basket:** it's really **gold-led**. AUDJPY is a marginal secondary; **AUDUSD (and untested NZDJPY) are not OOS-validated** and should be demo-only or dropped. The single-param-set basket EA is also suboptimal here (gold wants hold 11, AUDJPY wants 24) → **per-instrument params** would be the right enhancement.
- EA defaults set to the **gold-robust** set (StopATR 3 / MaxHoldDays 11 / ArmWindow 5).

► **Forward-OOS / demo plan (the real test — clock starts now)**

In-sample backtests (even OOS-split) can't fully clear overfitting; the honest validation is **forward**: 1. **Deploy on a demo account, gold-led**. Attach `RedditMacro_EA` to **XAUUSD** (and AUDJPY) on a demo, `StrategyTag=retail_fear`, ATR sizing 1%/trade, gold-robust params. Enable WebRequest for the `/signals` URL. 2. **The `/signals` feed already serves the live trigger** (capitulation + VIX + fear\_z); the EA mirrors it. From today, every fill is genuine OOS. 3. **Track ~2-3 fear episodes** (a few months) before judging — the edge fires only a handful of times/year. 4. Re-evaluate vs the backtest PF/DD; promote to (small) real size only if forward ≈ backtest.

★★★ **RANDOM-ENTRY / BUY-AND-HOLD RECKONING (2026-06-03) — the gold "edge" was drift**

The ultimate null test ( `backtest/mt5_sim.py::run_benchmarks` ): does entering on **capitulation** dates beat entering on the **same number of RANDOM dates** (identical turn/stop/hold/sizing), and does it beat **buy-and-hold**?

| leg    | strategy net    | random-entry (1000×): median / 95th | percentile | buy-and-hold     |
|--------|-----------------|-------------------------------------|------------|------------------|
| GOLD   | +8.1% (PF 1.52) | +18.8% / +29.2%                     | 4th        | +240.6% (DD 21%) |
| AUDJPY | +6.2% (PF 1.47) | +3.4% / +9.6%                       | 76th       | +30.7% (DD 30%)  |

- **GOLD has NO timing edge — it is WORSE than random** (4th percentile; random beat it 96% of the time) and **buy-and-hold (+240%) crushes it**. The PF 1.52 / OOS 1.27 that survived the excess study, MT5, OOS *and* optimization was **just gold's uptrend**; the capitulation timing actively enters at worse-than-random moments (fear-spike volatility whipsaws the turn-entry + stop). **The "one deployable edge" does not survive this null.**
- **AUDJPY**: 76th percentile — a *weak* positive tilt (better than median random) but **not significant**, and it doesn't beat its own buy-and-hold.
- **Lesson**: the per-trade excess study (which subtracts unconditional drift) *approximated* this null but UNDER-caught it once realistic mechanics (stop/turn/sizing) were added — the mechanics interact with the signal's volatility regime. **Random-entry + buy-and-hold are the non-negotiable final gate.**
- **Net effect on the macro side**: after this test there is **no robustly-deployable macro edge**.

✓ `distribution_short` **PASSES the same null (the contrast that matters)**

Ran the identical random-entry null on `distribution_short` ( `backtest/microstructure.py::random_null` ):

| test                                  | random median / 95th | dist_short percentile         |
|---------------------------------------|----------------------|-------------------------------|
| actual mean short net = +7.38% (n=62) | —                    | —                             |
| Null A — random dates, SAME tickers   | +0.02% / +3.87%      | <b>100th</b> (beats all 1000) |
| Null B — random concentrated pumps    | -1.36% / +5.86%      | <b>98th</b>                   |

- **The opposite of gold**. Shorting the same tickers on *random* dates → ~0%; shorting *random* concentrated pumps → -1.4%. `distribution_short`'s +7.38% sits at the **100th / 98th percentile** → both its **timing** (the price-rolling-over divergence) AND its **selection** (concentrated-pump + euphoric-sentiment gate) add real, non-drift value. **This is the one signal that earns the "real edge" label.**
- **Remaining caveats (so it's not over-trusted)**: single-regime (~2025-26, no historical penny prices to multi-regime validate); it **fades in its 2nd half** (date-split t=3.16→0.71); small-cap shorts have squeeze/ borrow/HTB tails → **express via puts**. So: *real, but not yet deployable-with-confidence* — needs forward OOS.

## Exhaustive null across EVERY strategy ( `backtest/null_all.py` ) — and the null's own limitation

Ran the random-entry null on all strategies. **Definitive KILLS (worse than random — the null is decisive here):** `organic_long` (1st pct), `dump_fade_short` (2nd), `deletion_short` (0th), and **every macro capitulation leg** (gold 4th, US500/USTEC 0th, silver 29th, AUDJPY 38th). **Methodological catch:** a naive broad/timing null **over-credits any strategy that enters near a big move** — `pump_long` (100th) "passes" only because it *longs the pump*, yet it regime-flips ( $t=0.75$ ); `attention_fade` / `fade_organic_short` likewise (tiny +0.3% beat-random- by-selection edges that flip across regimes). So **the null is necessary, not sufficient** — pair it with regime-stability + an *own-universe* control. Only **distribution\_short** **passes all of it** (timing null 100th, own-universe null 98th, AND it predicts a *reversal* rather than riding a contemporaneous spike). `smallcap` / `biotech` pass but are single-regime; `AUDUSD` is 96th full-period yet fails OOS. **Bottom line unchanged: one robust edge ( `distribution_short` ); the macro capitulation complex is confirmed worse-than-random.**

## distribution\_short — clean-universe validation + forward-OOS log ( `backtest/dist_short_oos.py` )

Focusing on the one survivor surfaced a **universe-pollution bug that was halving the edge**. The raw signal set leaked four off-thesis categories — mega-caps (AAPL/NVDA/META/PLTR...), common-word false positives (LIVE/GOOD/TIME/SOAR/BOOM/PR/SI — the *word*, validated as a symbol), ETFs (IAU/IWM/QQQ), and crypto (ETH). Excluding them **a priori, by what the names are (not their returns)**:

| universe                        | n         | net            | win        | t           |
|---------------------------------|-----------|----------------|------------|-------------|
| ALL (polluted)                  | 52        | +9.46%         | 62%        | 3.13        |
| <b>CLEAN micro-cap only</b>     | <b>25</b> | <b>+21.46%</b> | <b>84%</b> | <b>4.10</b> |
| excluded (mega/word/ETF/crypto) | 27        | -1.65%         | 41%        | -1.72       |

The thesis predicts exactly this: **manufactured-pump distribution is a micro-cap phenomenon** — Reddit moves price only where liquidity is thin; on mega-caps/ETFs/crypto the same short *loses* (they mean-revert up). The excluded set is a genuine mix (PLTR +6%, INTC -8%, NVDA +2%) netting ~zero, so this is universe-definition, not cherry-picking. **The "recent decay" earlier flagged was a pollution artifact** — on the clean universe the late half is +18.99% (win 85%,  $t=2.77$ ) vs early +24.13% ( $t=3.06$ ), i.e. *stable*, not fading. This **supersedes** the date-split "fades to +2.5%,  $t=0.71$ " line in the optimization table below — that split was run on the polluted set.

**Forward-OOS log is now live (spec frozen 2026-06-03).** `dist_short_oos.py` records every fresh clean-universe signal and its realized 10d outcome as the scheduler scrapes; signals dated  $\geq$  freeze are genuine OOS (the spec couldn't be fit to them). 25 in-sample signals seeded (+21.46%,  $t=4.10$ ); the OOS record starts at zero and builds forward. **This is the one gate distribution\_short hasn't faced** — the whole sample still lives in a single ~12-month scraped window, so only forward accumulation can confirm it. Remaining caveats: fat left tail (worst trade JAGU -42% short squeeze → **express via puts**, not naked short); blocklist needs maintenance as new mega-caps / common-word tickers appear.

## ★★ Comprehensive optimization across ALL strategies — the overfitting reckoning (2026-06-03)

Applied the same protocol (comprehensive genetic IS-optimize 2018-22 → OOS-validate 2023-26, edge params swept, RiskPct fixed at 1%) to **every MT5-tradeable instrument**, and a date-split pseudo-OOS to the Python equity legs. **The result is unambiguous: optimisation overfits almost everywhere; only one edge survives.**



| strategy / instrument                                              | IS (optimised)            | OOS                            | verdict                                                                                       |
|--------------------------------------------------------------------|---------------------------|--------------------------------|-----------------------------------------------------------------------------------------------|
| capitulation <b>GOLD</b> (robust config, <i>not</i> max-optimised) | PF 1.46                   | <b>PF 1.27-1.46, DD 22-33%</b> | ✅ <b>the one deployable edge</b>                                                              |
| capitulation AUDJPY (optimised)                                    | PF 2.01                   | <b>PF 0.99</b>                 | X overfit — collapsed                                                                         |
| capitulation AUDUSD (optimised)                                    | PF 2.47                   | <b>PF 0.73</b>                 | X overfit — lost                                                                              |
| capitulation Silver (optimised)                                    | PF 2.85                   | <b>PF 0.26 (DD 115%)</b>       | X overfit — blew up                                                                           |
| capitulation US500 (optimised)                                     | PF 1.91                   | PF 1.37                        | ~ held but weak signal (excess t≈0.9)                                                         |
| capitulation USTEC                                                 | —                         | —                              | no usable result                                                                              |
| <b>distribution_short</b> (equity, date-split)                     | +11.4% (t=3.16, 1st half) | ~~+2.5% (t=0.71, 2nd half)~~   | <b>! SUPERSEDED</b> — fade was pollution; clean universe late-half +19.0% (t=2.77), see above |
| <b>smallcap_pump_fade</b> , biotech                                | strong                    | <b>untestable</b>              | single-regime (no penny price history)                                                        |

- **The tell was visible before OOS:** each instrument's in-sample "optimum" was a *different random-looking corner* (turn 0/1, stop 1-4, hold 7-17, Fixed/Kelly) — no consistent pattern = curve-fit. OOS confirmed it: IS PF 2-2.85 → **OOS PF 0.26-1.37**.
- **Only GOLD survives**, and crucially only with a config chosen for **robustness (a plateau: turn-entry, stop 3, hold 11)**, NOT the max-IS optimum (which would also overfit). **distribution\_short** is the only other thing with real signal, but it fades in its second half and can't be cleanly OOS-tested (single-regime).
- **Net of optimising everything:** the system has **one robustly-deployable edge (gold capitulation) + one promising-but-unvalidated equity short (distribution\_short)**. Everything else is overfit, single-regime, or dead. Optimisation did not create edges — it exposed which were real.

## Comprehensive optimization (all 7 params, gold IS 2018-22) — profit-max = a leverage trap

Swept **every return-impacting param** (TurnEntry, ArmWindow, StopATR, MaxHoldDays, AtrPeriod, SizingMethod, RiskPctPerTrade), genetic, max-profit. Result:

| config                                                         | PF   | DD%       | net      |
|----------------------------------------------------------------|------|-----------|----------|
| profit-max (turn OFF, stop 1.0, hold 5, <b>risk 2.0</b> ) — IS | 1.54 | <b>67</b> | +\$425k  |
| same — <b>OOS 23-26</b>                                        | 1.45 | <b>67</b> | +\$15.6k |
| robust (turn ON, stop 3, hold 11, risk 1.0) — OOS              | 1.27 | <b>22</b> | +\$1.2k  |

- **RiskPctPerTrade maxed out at the sweep ceiling (2.0)** — it's pure **leverage** (scales profit *and* DD ~linearly), so a profit-max just compounds it to a **67% drawdown**. Risk% is a *risk-tolerance dial, not an edge param* — exclude it from profit optimization.
- The profit-max also flipped to **turn=OFF + tight stop + quick exit** — a different "cut-losses-fast" mechanism that **held OOS (PF 1.45)** but at 67% DD (undeployable). **AtrPeriod** showed no strong effect.
- **Optimising for profit picks a high-leverage / high-DD corner**. The deployable answer is the **risk-adjusted** one: the robust config (turn entry, stop 3, hold 11, risk 1%) — ~same PF at **a third of the drawdown**. EA defaults left on robust.

## ⚠️ MT5 deployable-config report + honesty correction (2026-06-03)

Ran the current per-instrument config in the MT5 tester:

| config                                  | trades | win% | PF          | DD% |
|-----------------------------------------|--------|------|-------------|-----|
| Gold full 2018-26 (stop3/hold11/1.0%)   | 63     | 63.5 | 1.46        | 33  |
| Gold OOS 2023-26                        | 16     | 56.3 | <b>1.27</b> | 22  |
| AUDJPY full 2018-26 (stop2/hold24/0.5%) | 54     | 46.3 | 1.63        | 3.1 |
| AUDJPY OOS 2023-26                      | 14     | 50.0 | 1.25        | 2.3 |

**Correction:** earlier I quoted **gold OOS PF 1.96**; that used **ArmWindow=7**, whereas the shipped default is **ArmWindow=5**, which gives **PF 1.27** on the same OOS. My "ArmWindow  $\geq 3$  is irrelevant" claim held *in-sample* (turns land within  $\sim 3$  bars in 2018-22) but **NOT out-of-sample** (2023-26 turns sometimes take 4-7 bars). On just **16 OOS trades** this swings PF 1.27 $\rightarrow$ 1.96 — so the honest statement is **gold is OOS-profitable, PF  $\approx$  1.3-2.0, arm-sensitive on a tiny sample** (not a hard 1.96). Both legs are +ve IS *and* OOS; AUDJPY adds steady low-DD (2-3%) return as a half-size satellite. Forward demo remains the real test.

## ✅ Full backtest-suite refresh (2026-06-03, on the extended 2016-26 data, 100% pump-author coverage)

Re-ran every Python backtest module. **Conclusions held — the system is stable.**

| module / signal                       | result on current data                              | verdict                                                                         |
|---------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------|
| run_all retail_fear AUDJPY            | excess <b>+0.21pp</b> , win 62%                     | ✅ SURVIVOR (core macro edge)                                                    |
| run_all retail_fear SPY / QQQ         | excess +0.12 / +0.24pp, win 66%                     | ✅ marginal survivors (SPY flipped $-0.05 \rightarrow +0.12$ on the longer data) |
| microstructure distribution_short     | <b>+7.38%/10d, t=2.83</b> , +ve all 3 regimes       | ✅ holds (strengthened)                                                          |
| sectors smallcap_pump_fade            | +8.54% excess short (t= $-3.89$ )                   | 🟡 single-regime                                                                 |
| sectors biotech_catalyst              | +56% but win 42% (t=2.13)                           | 🟡 lottery                                                                       |
| macro_research euphoria/momentum/fade | all negative-excess except a marginal QQQ down-fade | ❌ no edge                                                                       |
| run_all attention_fade / organic_long | -1.21 / -3.74                                       | ❌ dead                                                                          |
| calendar_events turn-of-month         | +1.20%/5d, <b>t=6.46</b> (Nov, seasonality)         | 🟡 real but generic equity overlay, not Reddit alpha                             |
| trend AUDJPY trend-gate               | ungated robust every year; gating thins it          | confirms AUDJPY stays <b>ungated</b>                                            |
| trend euphoria_short                  | only +ve in the 2022 bear                           | confirms <b>parked</b>                                                          |

**Caveat (the gap that matters):** these are the *statistical* (excess-vs-B&H) edges; the macro excesses are thin (+0.1-0.2pp). The *tradeable* reality from the MT5 + OOS work is narrower — only **gold** survived realistic single-entry execution out-of-sample (PF 1.96/DD 18%), AUDJPY marginal, AUDUSD/NZDJPY failed. So: statistically real = capitulation-long + distribution\_short; realistically deployable = **gold-led**, forward-demo pending.

## Changelog

- **2026-06-03 — ★ "Two edges"  $\rightarrow$  ONE edge; the ROLLOVER gate is the engine (gate-ladder).** Decomposed distribution\_short vs smallcap\_pump\_fade on a common clean universe. Concentration alone is  $\sim 0$  and flips negative in 25-26 (smallcap core  $-0.22\%$ ; sentiment barely helps (+2.95%, t1.0); **the rollover gate (trailing-5d $<0$ )  $\sim 4\times$  the edge** ( $\rightarrow$  +12.69%, t4.11). smallcap\_pump\_fade's "robustness" was the curated sector-universe selection, not strategy logic  $\rightarrow$  **downgraded, not an independent edge**. Useful upgrade: a **capacity knob** — looser concentration (pa $\geq 3$ , m $\geq 8$ ) + the same gates gives n=81 (+17% signals), +9.5%, and positive in EVERY regime (the high-conviction pa $\geq 2$ , m $\geq 10$  variant is +12.7% but flat in 2021).

- **2026-06-03 — ★ FULL revalidation audit on the multi-year corpus (564k mentions, 4-5× data).** Re-ran `null_all` + regime-split + macro null + distribution\_short on the expanded 2021-2026 data. **All kills held** (macro capitulation 3rd-pct drift; organic\_long/dump\_fade/deletion worse-than-random). The null's "REAL EDGE" labels were again exposed by the regime-split: `biotech_catalyst` FLIPS (-8% 2021 → +22% now), `attention_fade` & `fade_organic_short` are **one-period (2024-only)** artifacts, `pump_long` is a **regime LOSER** (-2/-10/-19/-12% in 2021-24). **One upgrade:** `smallcap_pump_fade` is **sign-stable positive across every regime** (2021 +6% / 2024 +4% / 25-26 +10%, t4.2) — a broader/weaker cousin of `distribution_short`, the only "other" strategy to survive the regime-split → worth investigating the union. `distribution_short` re-confirmed (+11.78%, null 100th, robust via puts).
- **2026-06-03 — ★ Multi-regime validation + frothy-retail regime added to the scoreboard.** Deep-backfilled the corpus to 2021 ( `archive_backfill_deep` , 5 micro-cap subs, 330k+ posts; DuckDB research store, 9.6× faster). Three-regime test ( `backtest/regime_test.py` , `backtest/distribution_short.py` ): `distribution_short` is **regime-CONDITIONAL** — 2021 mania null 46th (DEAD, re-squeezes), 2022 bear +18.7%, 2025-26 +20.96% (null 100th). **MT5-validated on broker OHLC** ( `scripts/mt5_dist_short_backtest.py` ): 36/37 names tradeable, +11.02%, pattern replicates. **Cap-tier:** works micro→small→mid, reverses on large/mega; strongest in liquid \$5-15 band (t=5.26). Added **frothy-retail-regime** as a tracked regime-filter idea (detector unbuilt — froth-index failed).
- **2026-06-03 — ★ distribution\_short clean-universe + forward-OOS log ( `backtest/dist_short_oos.py` ).** Found the signal universe was polluted with mega-caps / common-word false-positive tickers / ETFs / crypto — excluding them a priori **doubled the edge to +21.46% (win 84%, t=4.10)** and revealed the earlier "recent decay" was a pollution artifact (clean late-half +19.0%, t=2.77 — stable). Froze the clean spec and started the forward-OOS log: every fresh signal ≥ 2026-06-03 is genuine out-of-sample, accumulating as the scheduler scrapes. The last untested gate is now ticking.
- **2026-06-03 — Exhaustive random-entry null on ALL strategies ( `backtest/null_all.py` ).** Definitive kills (worse than random): `organic_long`, `dump_fade_short`, `deletion_short`, and every macro capitulation leg (gold 4th, US500/USTEC 0th). Key methodological finding: the naive null **over-credits momentum/selection strategies** (`pump_long` 100th but regime-flips; `attention_fade/fade_organic` 99th but flip) — it's necessary, not sufficient; must pair with regime-stability + own-universe control. Only **distribution\_short passes everything**. Bottom line unchanged: one robust edge, macro capitulation confirmed worse-than-random.
- **2026-06-03 — distribution\_short PASSES the random-entry null — the one real edge.** Same null applied ( `microstructure.random_null` ): vs random dates on the same tickers → **100th percentile**; vs random concentrated pumps → **98th**. So its timing (divergence) AND selection (gate) add real, non-drift value — the exact opposite of gold (4th pct). Still single-regime + fades in 2nd half + small-cap → puts, so real but not yet deployable-with-confidence. The system's honest state: **ONE real edge (distribution\_short), no deployable macro edge**.
- **2026-06-03 — ★ RANDOM-ENTRY RECKONING — the gold edge was drift.** Added random-entry Monte Carlo (1000×, same #entries + mechanics, random dates) + buy-and-hold to `mt5_sim`. **GOLD capitulation is at the 4th percentile of random** (worse than random; B&H +240% vs strategy +8%) → the PF that survived excess/MT5/ OOS/optimization was just **gold's uptrend**, with the signal timing it *worse than random*. AUDJPY only 76th (weak, n.s.). **No deployable macro edge survives this null** — `retail_fear` downgraded to ❌. `distribution_short` (has a built-in control) is the only signal with a null-beating result, but is single-regime. Random-entry + B&H are now the mandatory final gate.
- **2026-06-03 — Full backtest suite + optimization now runs LOCALLY in Python.** Added a local full-grid optimizer to `mt5_sim` (IS→OOS, on the MT5 broker OHLC, ~7k sims in seconds) + series caching. Re-ran everything locally: the MT5-replica reproduces the reckoning (only GOLD holds OOS, max-IS PF 2.19; AUDJPY/Silver/US500 overfit→fail), and the Python suite is unchanged (`capitulation_long` SURVIVOR SPY/QQQ/AUDJPY, `distribution_short` +7.4% t=2.83, smallcap/biotech single-regime, turn-of-month overlay). No MT5 dependency for backtesting any more — MT5 is just the live execution shell.
- **2026-06-03 — Pulled MT5 broker OHLC into Python → the replica now MATCHES the MT5 tester.** Used the `MetaTrader5` Python package ( `scripts/export_mt5_ohlc.py` ) to export D1 OHLC straight from the terminal (the exact tester data) for all 6 instruments → `mt5_ohlc` table. `mt5_sim.py` now uses true-range ATR + intraday stops and **reproduces MT5** (gold OOS PF **1.27 == 1.27**; AUDUSD/silver fail OOS in both). The strategy now runs fully in Python on the real broker data, MT5-free. Residual gaps: closed-trade vs floating-equity DD, and no spread/swap (FX) — both addressable.
- **2026-06-03 — Python replica of the MT5 EA ( `backtest/mt5_sim.py` ) — cross-check.** Re-ran the capitulation legs in Python (turn entry / ATR stop / risk sizing). Qualitative verdicts reproduce (gold +ve, **silver overfits 2.02→0.76 OOS**, all legs thin PF ~1.0-1.5) — BUT the prices table is **close-only**, so DD is badly understated (Python 4-7% vs MT5 22-67%) and stops are approximate. Confirms the edge story; NOT reliable for drawdown. **Data gap to close: OHLC prices** would make the Python backtest MT5-accurate.
- **2026-06-03 — Optimized ALL strategies (IS→OOS) → the overfitting reckoning.** Comprehensive genetic optimization of capitulation-long on every MT5 instrument (AUDJPY/AUDUSD/Silver/US500/USTEC) + date-split for `distribution_short`. **Every per-instrument optimum overfit and failed OOS** (IS PF 2.0-2.85 → OOS 0.26-1.37); each "best" was a different curve-fit corner.

- Only GOLD survives** (robust config, OOS PF 1.27-1.46). `distribution_short` real but fades (1st-half  $t=3.16 \rightarrow$  2nd-half  $t=0.71$ ). smallcap/biotech untestable (single-regime). Conclusion: one deployable edge (gold) + one unvalidated (`distribution_short`).
- **2026-06-03 — Comprehensive optimization (all 7 params) → profit-max is a leverage trap.** Swept `TurnEntry/ArmWindow/StopATR/MaxHoldDays/AtrPeriod/SizingMethod/RiskPct` (genetic, gold IS). `RiskPct` maxed to the 2.0 ceiling (it's leverage, not edge) → 67% DD; profit-max flipped to turn-OFF/tight-stop/quick-exit (+\$425k IS, held OOS PF 1.45 but 67% DD = undeployable). The risk-adjusted answer stays the robust config (`turn/stop3/hold11/risk1%`): ~same PF at 22-33% DD. Lesson: optimise risk-adjusted, set `RiskPct` by tolerance.
  - **2026-06-03 — MT5 deployable-config report + honesty correction.** Ran the current per-instrument config: Gold full PF 1.46 / OOS 1.27 (DD 22-33%); AUDJPY full PF 1.63 / OOS 1.25 (DD 2-3%). Both legs +ve IS *and* OOS. **Corrected an overstatement:** the earlier "gold OOS PF 1.96" used `ArmWindow=7`; the shipped default `arm=5` gives PF 1.27 on the same 16 OOS trades. "`ArmWindow≥3` irrelevant" held IS but NOT OOS — so gold OOS is PF  $\approx 1.3-2.0$ , arm-sensitive on a tiny sample (not a hard 1.96).
  - **2026-06-03 — Full backtest-suite refresh.** Re-ran `run_all / fear_gate / microstructure / sectors / calendar_events / trend / macro_research` on the extended 2016-26 data. All conclusions held: capitulation-long a consistent SURVIVOR (AUDJPY +0.21 / SPY +0.12 / QQQ +0.24 excess), `distribution_short` +7.38%  $t=2.83$ , smallcap/biotech single-regime, euphoria/momentum/attention dead, turn-of-month a generic overlay. System stable.
  - **2026-06-03 — Gold-led basket + per-instrument EA params.** Trimmed live `INSTRUMENTS` to the OOS survivors: **XAUUSD (primary) + AUDJPY (small secondary)**; dropped AUDUSD (OOS PF 0.47) and NZDJPY (no OOS). EA live basket now uses **per-leg** `StopATR/MaxHoldDays/Risk` (gold 3/11/1.0%, AUDJPY 2/24/0.5%) instead of one shared set — so each instrument runs at its own OOS optimum. Recompiled; tester path unchanged (gold full-period PF 1.46 sanity-checked).
  - **2026-06-03 — Optimized AUDJPY+AUDUSD too → only GOLD survives OOS; set gold defaults; forward plan.** Same IS→OOS protocol: Gold robust (OOS PF 1.96/DD 18%), AUDJPY marginal (OOS PF 1.31, needs  $\text{hold} \approx 24$ , default `hold=10` goes -ve), **AUDUSD overfits & fails OOS (1.78→0.47)** — its +excess  $t$ -stat doesn't trade (it declined). So the live basket is **gold-led**; AUDUSD/NZDJPY demo-only/drop. Set EA defaults to the gold-robust set (`StopATR 3/hold 11/arm 5`, recompiled). Added a forward-OOS/demo deployment plan.
  - **2026-06-03 — MT5 optimizer run (gold): in-sample optimize → OOS validate.** Genetic sweep of entry/exit params, profit-maximised IS (2018-22), validated OOS (2023-26). Signal didn't badly overfit (all sets OOS-profitable, PF 1.5-2.0) but max-profit = a high-DD corner (42-55%); the **robust set (stop 3×ATR / hold 11) is best OOS: PF 1.96, DD 18%, win 62%**. Added headless frame-logging + `RedditMacro_Optimize.ini`.
  - **2026-06-03 — Turn entry + stop makes it tradeable; sizing methods backtested.** Added to the EA: arm on capitulation → enter on the first up-bar (turn) within `ArmWindow`, with a `StopATR×ATR` stop; plus selectable sizing (`Fixed/ATR/Kelly`). The turn entry **rescues the strategy**: AUDJPY PF 0.77→1.06, Gold PF 1.25→1.54 (Sharpe 0.41). Sizing: `ATR/Fixed` best; **Kelly oversizes** (caps out on thin edges → 30-80% DD, no return gain). Gold is the strongest leg. Still in-sample/preliminary → forward OOS + demo next.
  - **2026-06-03 — MT5 reality check: excess edge ≠ tradeable.** Fixed the EA (`ATR` fixed-fractional sizing trading-day holds; compiled clean), which collapsed the fixed-lot drawdown artifact (90%→17-20% on FX) and revealed the truth: under realistic single-entry execution the capitulation longs are breakeven-to- negative (AUDJPY PF 0.85, AUDUSD 0.91, gold 1.01 but 93% DD). The validated *excess* edge doesn't survive naive long-on-trigger execution — gap is entry timing (`cluster-start` = falling knife) + `excess≠absolute` + no stop. Needs a turn-based entry + stop to deploy. Basket downgraded to research-validated, not live-tradeable.
  - **2026-06-03 — Trimmed live basket to tradable-alpha only.** Per the cross-asset map, kept the four legs with significant excess ( $t \geq 2$ ): **AUDJPY (2.93), NZDJPY (3.18), AUDUSD (2.72), Gold (2.03)**. Dropped US500/USTEC ( $t \approx 1$ ) and Silver (1.72, lumpy); parked `euphoria_short` (traded the dropped indices, -1.84 excess). `signals.py INSTRUMENTS` + MT5 EA basket trimmed 7→4; `GATE_TREND` cleared. Deployed live.
  - **2026-06-03 — Cross-asset generalization map + MT5 FX backtest.** Capitulation-long tested across FX/indices/commodities: it's a **risk-appetite/fear-reversal factor** — strong in risk-on FX (AUD/NZD, EM-ish), works in gold/silver, **flat in equity indices (incl. ASX  $t=0.08$ ) and growth commodities (copper/oil)**; AUDNZD control flat. MT5 tester (D1, 2016-26, ungated): AUDJPY +\$10.4k/PF 1.14, AUDUSD ~flat-\$ (declined, +excess only), 60% DD → needs VIX gate + vol-target to deploy.
  - **2026-06-03 — Cross-sectional FX validation → broadened the basket.** Tested capitulation-long across the yen-cross + risk-pair complex (2018-26, VIX-gated). The edge is a **risk-on-currency factor**: every AUD/NZD pair is significant (NZDJPY  $t=3.18$ , AUDJPY 2.93, AUDCHF 2.83, AUDUSD 2.72) *independent of the funding leg*, while the no-risk-leg controls are flat (USDJPY  $t=1.02$ , CHFJPY 1.37, EURUSD 1.44) — the controls failing is the proof it's real, not data-mining. It's NOT a JPY thing (JPY is just a clean funding leg; Nikkei works as high-beta risk, not as "Japanese"). **Broadened the live retail\_fear FX basket to AUDJPY + NZDJPY + AUDUSD** (`signals.py` + MT5 EA).
  - **2026-06-03 — Deletion short DIED at full coverage; AUDJPY re-confirmed; sweep de-crypto'd.** (1) Targeted-profiled all 487 equity pump-authors (25%→100% relevant coverage). The deletion short **flipped negative** — gone  $\geq 0.20 \rightarrow -19.3\%/10d$  ( $n=87$ ), `young_frac` short -39%→-87% — dominated by squeeze tails (IXHL -625%) the 25% sample had missed. The +14.3% was a **coverage-bias artifact**. Dead as a stock short; puts-only. `distribution_short` (price-rolling-over filter) is the survivor.

- (2) **AUDJPY capitulation re-validated:** VIX-gated +0.48%/10d **t(exc)=2.93 (2018-26)**, the gate excludes calm 2016-17), **+ve 8/9 years**; ungated +0.26% t=2.52 (2016-26) (2018 the lone falling-knife miss). Now the most significant edge in the system; unaffected by author work (pure macro). (3) `get_authors_needing_age` now skips crypto-only authors + front-loads ticker-mentioners.
- **2026-06-03 — Crypto leading-indicator test: COINCIDENT, not leading.** Price lead-lag BTC↔SPY ≈0 (same-day +0.17 but k≠0 noise), weekend test t=1.2, crypto sentiment coincident with equity RRAI (peaks k=0). Combined crypto+equity RRAI does NOT improve capitulation — *dilutes* the AUDJPY edge (+0.55%→+0.45%). Crypto's value = its own pump/meme universe, not macro timing. (See "Crypto — coincident" section.)
  - **2026-06-03 — RRAI extended to 2016-06 (525 caps, +190) → fear-gate STRENGTHENED.** Backfill landed (aggregate\_daily 2016-06→2026, 3,653 days; +2018 Volmageddon/Q4 episodes). Re-ran `backtest.fear_gate` : fear-only excess grew SPY +0.32→+0.57, QQQ +0.41→+0.80 (effect *grew* OOS = real-edge signature); episode-level now **significant on BOTH legs and they agree** (SPY t=2.77, QQQ t=2.37, k=63) — the prior SPY≠QQQ noise-floor flag is resolved. Day-level still marginal (1.6-1.9). The LIVE `trend+fear` gate is still ≈0 (SPY +0.10/QQQ +0.11) ⇒ **drop the trend filter, use fear-only**. AUDJPY confirmed best ungated.
  - **2026-06-02 — Power analysis + RRAI history extension (in progress).** Quantified the fear-gate's data need: the 94 fear-days are ~40 independent episodes; day-level it's t≈0.6 (needs ~9× more obs for p<0.05, ~19× for power — impractical), but episode-level it's t=2.28 (SPY)/1.85 (QQQ) — significant-but-fragile (COVID-cluster-driven; SPY≠QQQ ⇒ noise floor). Cheapest lever = more *independent fear episodes* via **backward** history (Wikipedia fear data already reaches 2015). Extended study prices to max and **kicked off the RRAI aggregate backfill 2016-06→2020** (wallstreetbets/stocks/pennystocks) to add the 2018 Volmageddon + Q4-2018 episodes; will re-run `backtest.fear_gate` on the longer series when it lands.
  - **2026-06-02 — Committed EXCESS study settles the fear-gate ( `backtest/fear_gate.py` ).** The proper frame (mean excess vs buy-&-hold, the +0.36pp's own metric). Three findings: (1) the +0.36pp **reproduces** as a point estimate (SPY +0.32 / QQQ +0.41 fear-only excess) — not fabricated; (2) it's **not significant** (excess t≈0.6, n≈94); (3) the LIVE **trend+fear** combo is *worse* — dilutes SPY (→+0.09), flips QQQ negative (→-0.20) — because the trend filter strips the downtrend bounces that carry the excess. Reconciles the MT5 "loss": MT5 sums absolute P&L (COVID-cluster dominates), excess means-averages — so the earlier "drag" was a frame artifact. Gate stays 🟡; fear-only > trend+fear for excess; AUDJPY stays ungated (the only significant edge). Exporter refactored to share the study's `capitulation_sets()` (MT5 + Python test identical dates).
  - **2026-06-02 — MT5 backtest of the fear-gate → it's a DRAG, not a rescue.** Extended `--export-signals` to emit 4 point-in-time capitulation date-sets (baseline / `fear_z≥0.5` / 200-DMA-trend / trend+fear) and ran each through the MT5 Strategy Tester on US500 (2020-26) via `RedditMacro_US500*.ini`. Adding `fear_z` made absolute returns *worse* (alone -\$1045, with trend -\$372 vs trend-only +\$21) — fear spikes fire long into tops (Feb-2020). Only the trend filter helps (DD 13.6→7.1%). Caveat: MT5 = absolute long-only, the +0.36pp was excess — so not a direct refutation, but the gate is downgraded to 🟡 doubtful and needs a committed *excess-based* study to settle (maybe drop `fear_z`, keep trend).
  - **2026-06-02 — Doc-completeness pass.** Audited the doc against the live registry + all backtest modules. Found one missing edge — the **Wikipedia fear-confirmation gate** (live: gates the equity legs of `retail_fear`, `fear_z≥0.5`, +0.36pp rescue) — now documented, *honestly flagged as not yet committed-backtested*. Also added the full live-screen list ( `fresh_pump_alert`, `coordination_flag`, ...) to the scoreboard and recorded the broad "attention-leads-price" long as a failed up-leg test.
  - **2026-06-02 — New PIT-tradeable short found: `distribution_short` ( `backtest/microstructure.py` ).** Built the missing **new-author recruitment** feature and tested demand-quality timing signals. The decisive lesson: microstructure timing (recruitment-exhaustion / deceleration / divergence) is **worthless on the broad attention universe** (it drifts flat-to-up, costs dominate) but **powerful inside the concentrated/manufactured subset** — exactly where the deletion edge lives. Winner = concentrated pump + euphoric sentiment + price already rolling over → **+7.2%/10d net, t=2.70 (date-clustered t=2.52, 47 indep. dates), dose-response, survives 2× costs**; control loses -4.1%. Unlike the deletion gate it uses **no hindsight** → tradeable. Recorded negatives (broad timing, recruitment-alone, up-leg ride) to avoid re-runs.
  - **2026-06-02 — Deletion-gated short forensically validated.** Author coverage 8%→20% (2,973 gone). Re-backtest: `gone_frac≥0.20` → +14.3%/10d short (83% win, t=2.57, +ve all regimes); the *control* (un-gated concentrated pumps) *loses* -10.8% — so the deletion gate flips the trade, confirming the A2 thesis. **Caveat:** `gone_frac` is hindsight (no Reddit deletion timestamp) → not yet PIT-tradeable; the `young_frac` PIT proxy is still too sparse (n=3-7). Also shipped `scraper/finra.py` (FINRA Reg SHO daily short-volume, 475k rows) + z-score `squeeze_screen()` — learned the *absolute* short-ratio is market-maker noise (median 0.49), so the screen uses per-ticker z-score anomaly + chatter.
  - **2026-06-01 — Instrument scan (17 MT5 instruments).** `retail_fear` generalises across the risk-on basket; commodities express it best (Silver +2.2pp lumpy, Gold +0.6pp robust), AUDJPY steadiest. Oil/crypto dropped. Basket = AUDJPY + Gold + indices + small Silver; EA + feed updated.
  - **2026-06-01 — Macro R&D + MT5 framework.** Benchmarked candidates vs buy-and-hold: only the `retail_fear` contrarian family beats it (best AUDJPY +0.47pp/10d excess); momentum/froth = no edge, all shorts fail. Built `SignalClient.mqh` + `RedditMacro_EA.mq5` (strategy-selectable), multi-strategy `/signals` feed; retired `RedditRRAI_EA.mq5`.

- **2026-06-01** — Initial document. Phase 0 (feature store + friction-aware backtester) shipped. RRAI capitulation-long validated (best via AUDJPY, VIX-gated); pump-ride/naive-dump-short/organic-long failed; fade-organic marginal; deletion-gated short awaiting author coverage (8%). `/signals` + MT5 EA live.